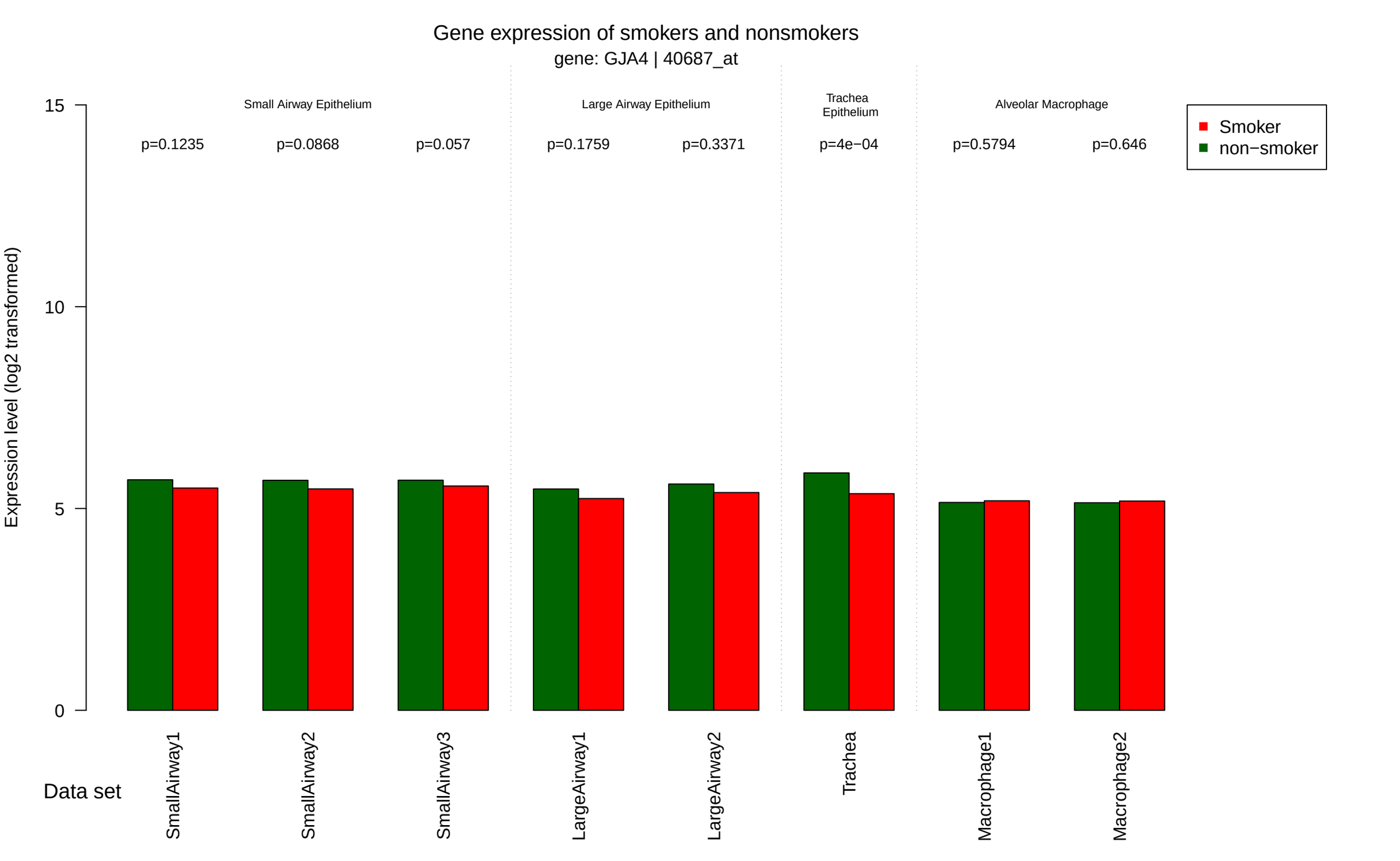
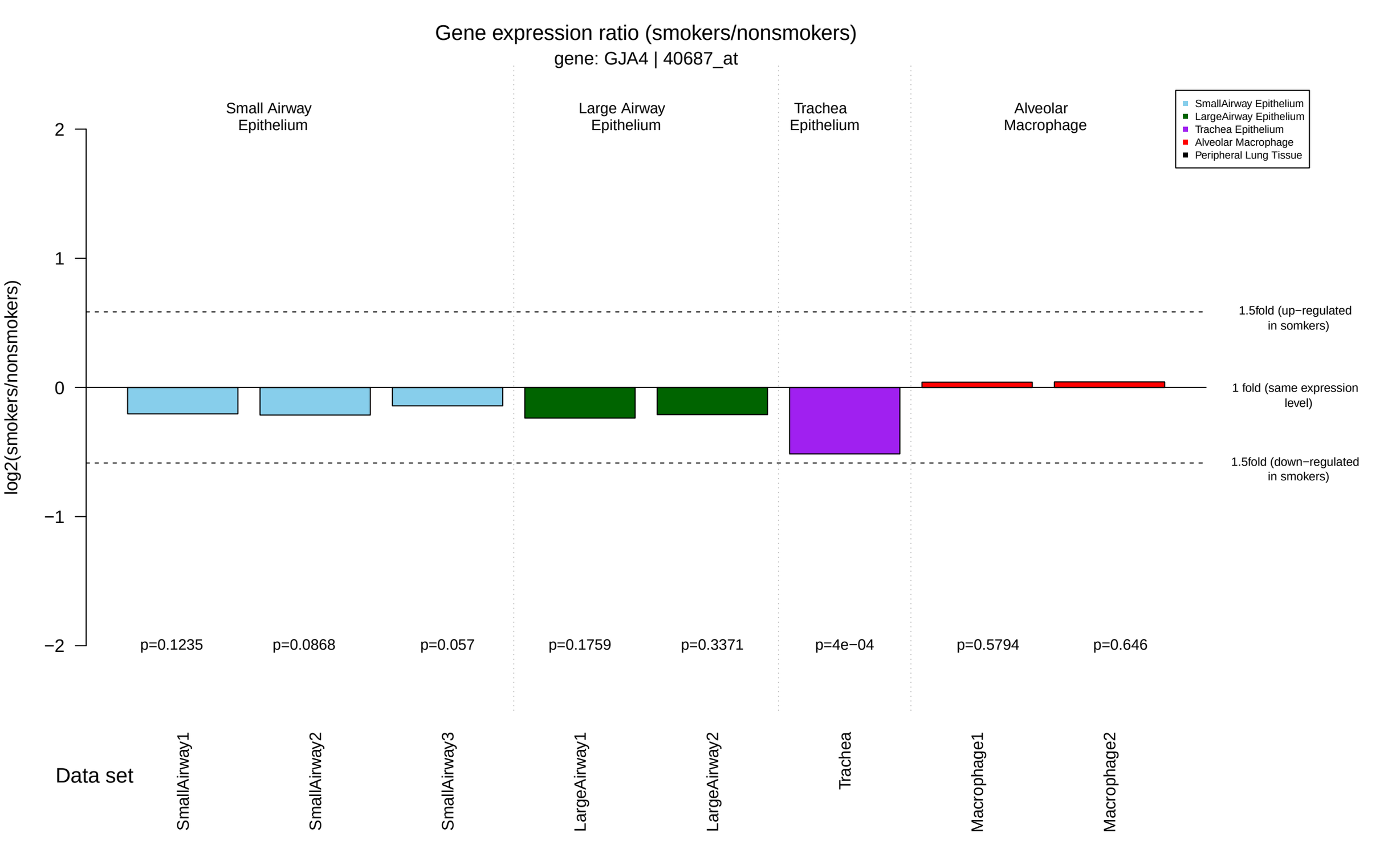
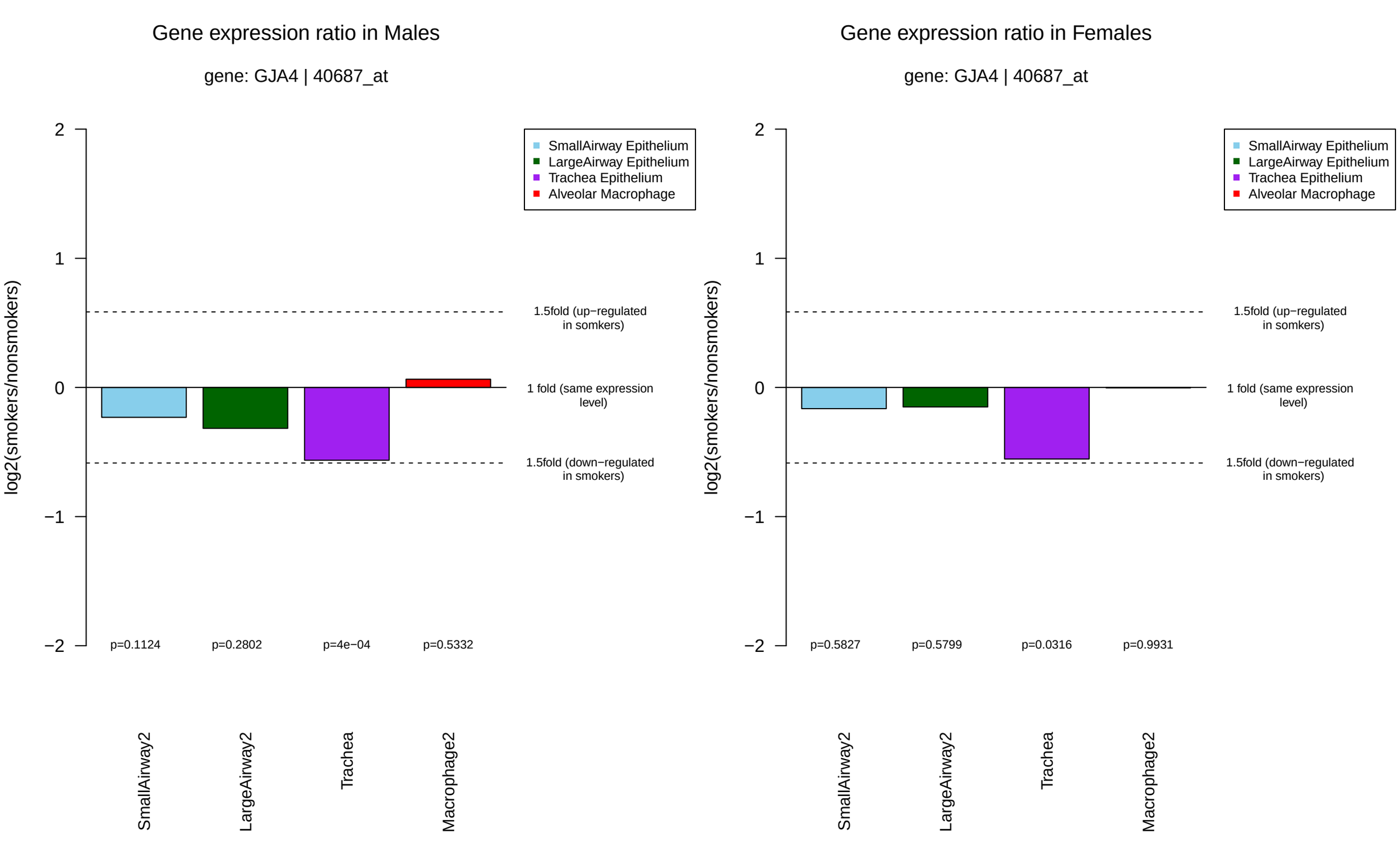
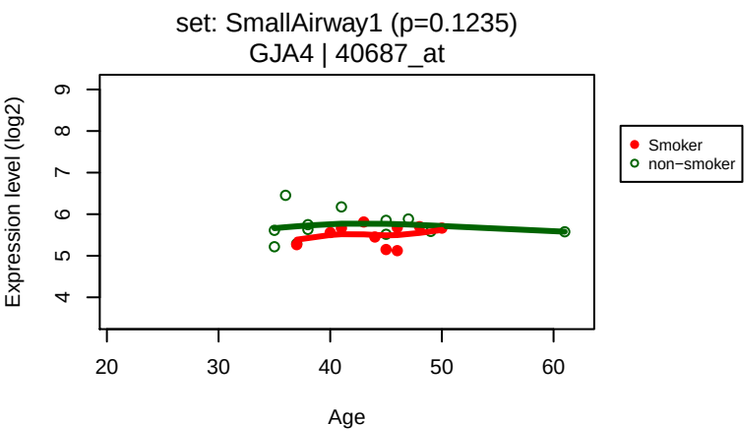
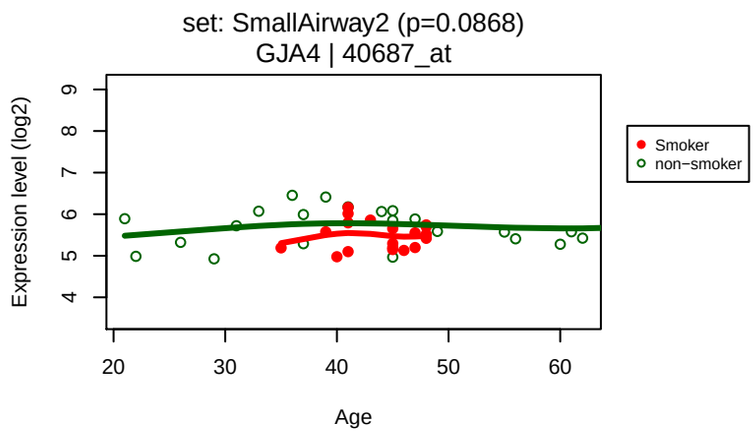
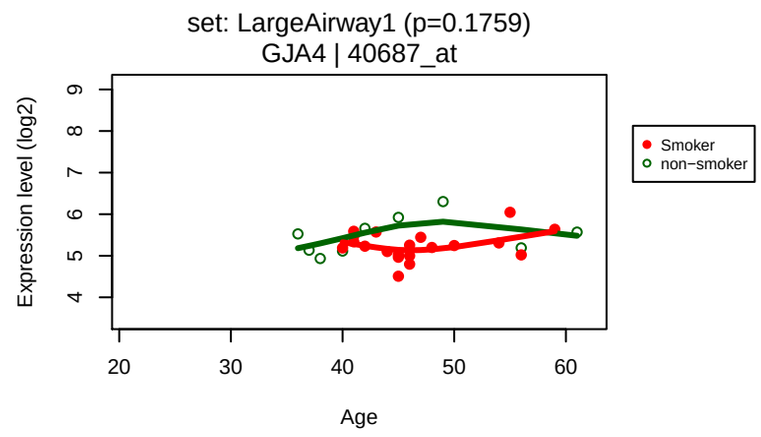

Gene expression of smokers and nonsmokers
gene: GJA4 | 40687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1235
p=0.0868
p=0.057
p=0.1759
p=0.3371
p=4e−04
p=0.5794
p=0.646
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

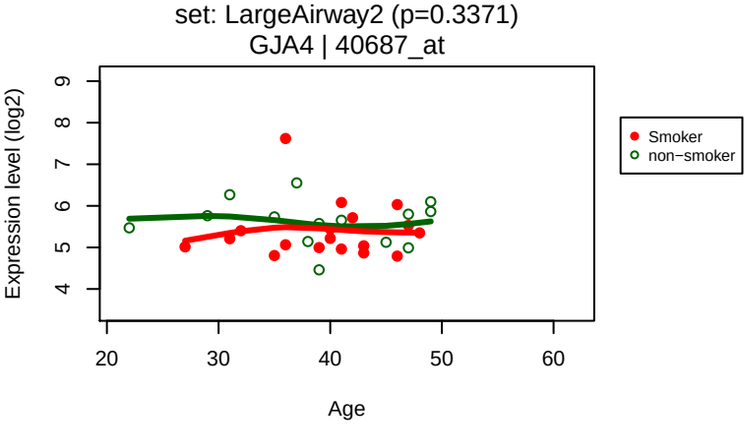

Gene expression ratio (smokers/nonsmokers)
gene: GJA4 | 40687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1235
p=0.0868
p=0.057
p=0.1759
p=0.3371
p=4e−04
p=0.5794
p=0.646
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

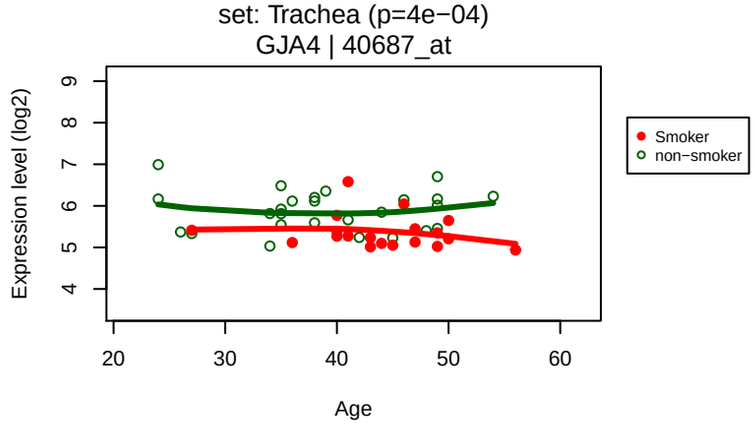

Gene expression ratio in Males
gene: GJA4 | 40687_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1124
p=0.2802
p=4e−04
p=0.5332
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GJA4 | 40687_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5827
p=0.5799
p=0.0316
p=0.9931
SmallAirway2
LargeAirway2
Trachea
Macrophage2

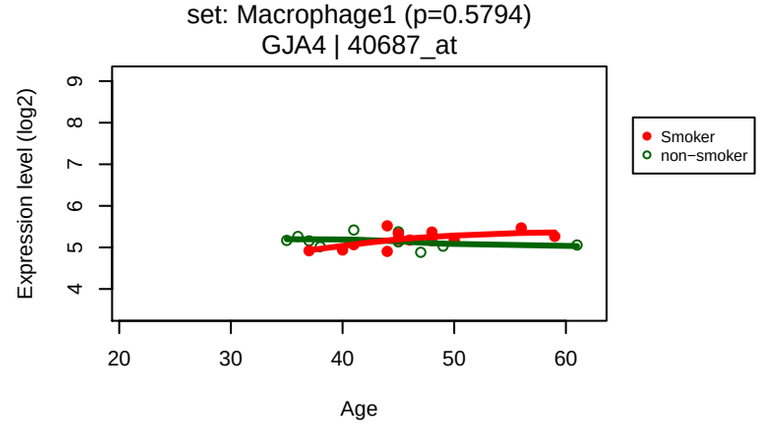

set: SmallAirway1 (p=0.1235)
GJA4 | 40687_at
Expression level (log2)
Age
Smoker
non−smoker

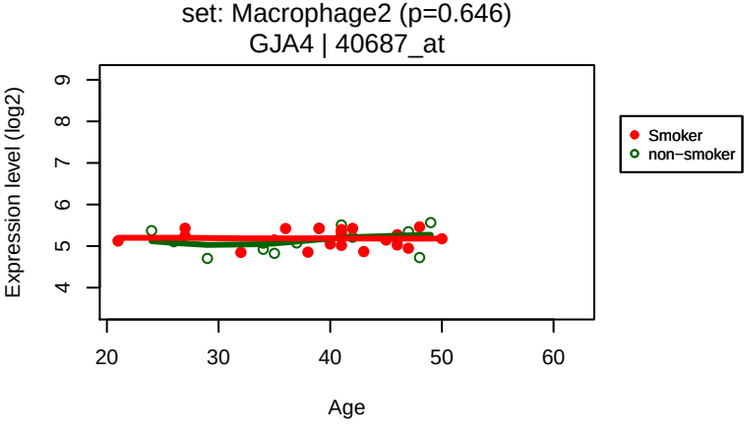

set: SmallAirway2 (p=0.0868)
GJA4 | 40687_at
Expression level (log2)
Age
Smoker
non−smoker

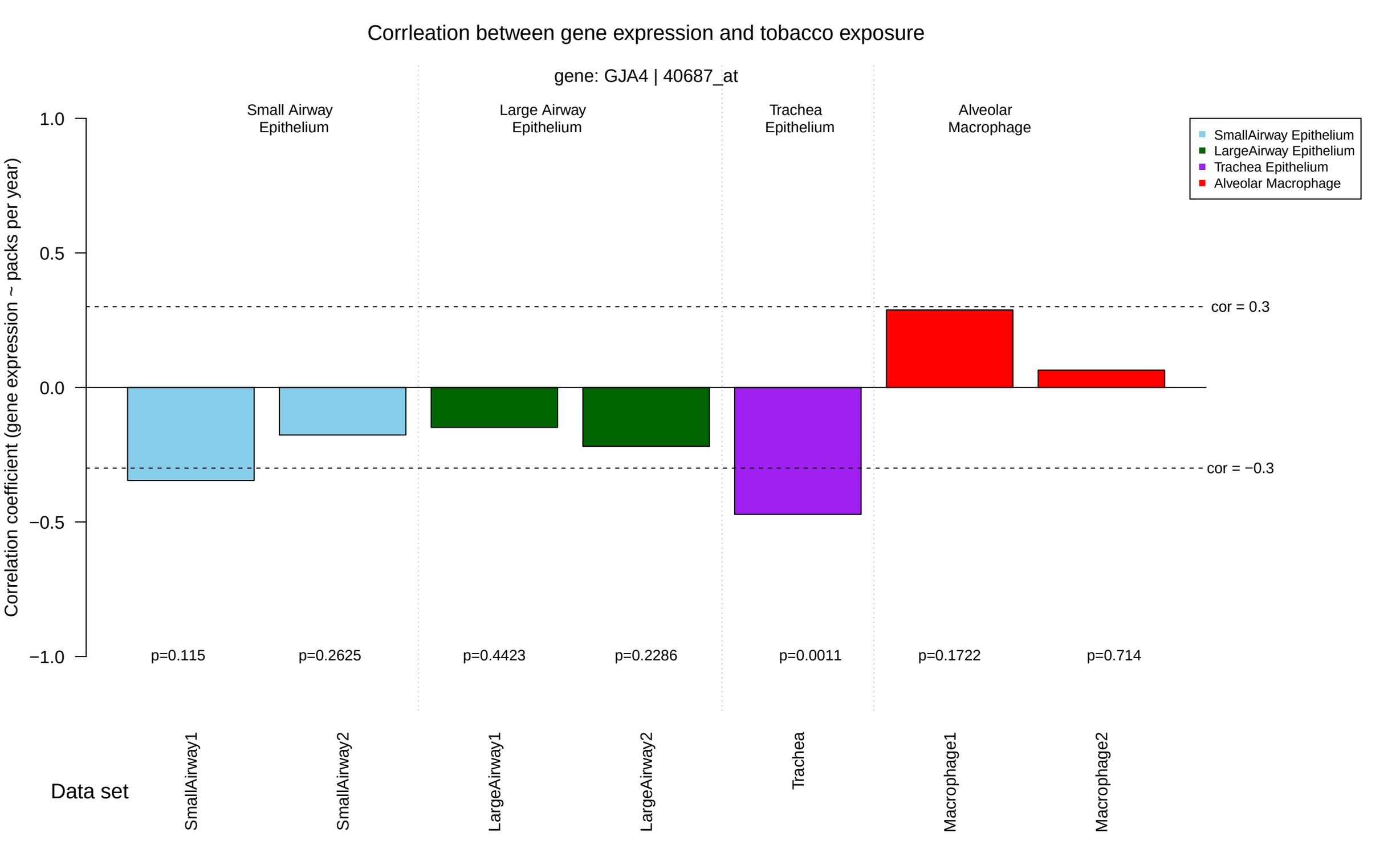

set: LargeAirway1 (p=0.1759)
GJA4 | 40687_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3371)
GJA4 | 40687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
GJA4 | 40687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5794)
GJA4 | 40687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.646)
GJA4 | 40687_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GJA4 | 40687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.115
p=0.2625
p=0.4423
p=0.2286
p=0.0011
p=0.1722
p=0.714
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2